S3 — Equilibrium and Acid-Base Chemistry

3.1 Dynamic Equilibrium

Dynamic equilibrium: A state reached in a closed system when the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant but are not necessarily equal. Both forward and reverse reactions continue to occur at the molecular level.

Conditions for dynamic equilibrium:

- The system must be closed (no substances can enter or leave).
- The forward and reverse rates must be equal.
- The concentrations of all species remain constant over time.
- The reaction must be reversible.

Le Chatelier's Principle

Le Chatelier's principle: If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium will shift in the direction that tends to oppose the change, so as to partially restore the original conditions.

Change	Effect on equilibrium position	Effect on K
Increase concentration of a reactant	Shifts to the right (towards products)	No change
Decrease concentration of a product	Shifts to the right	No change

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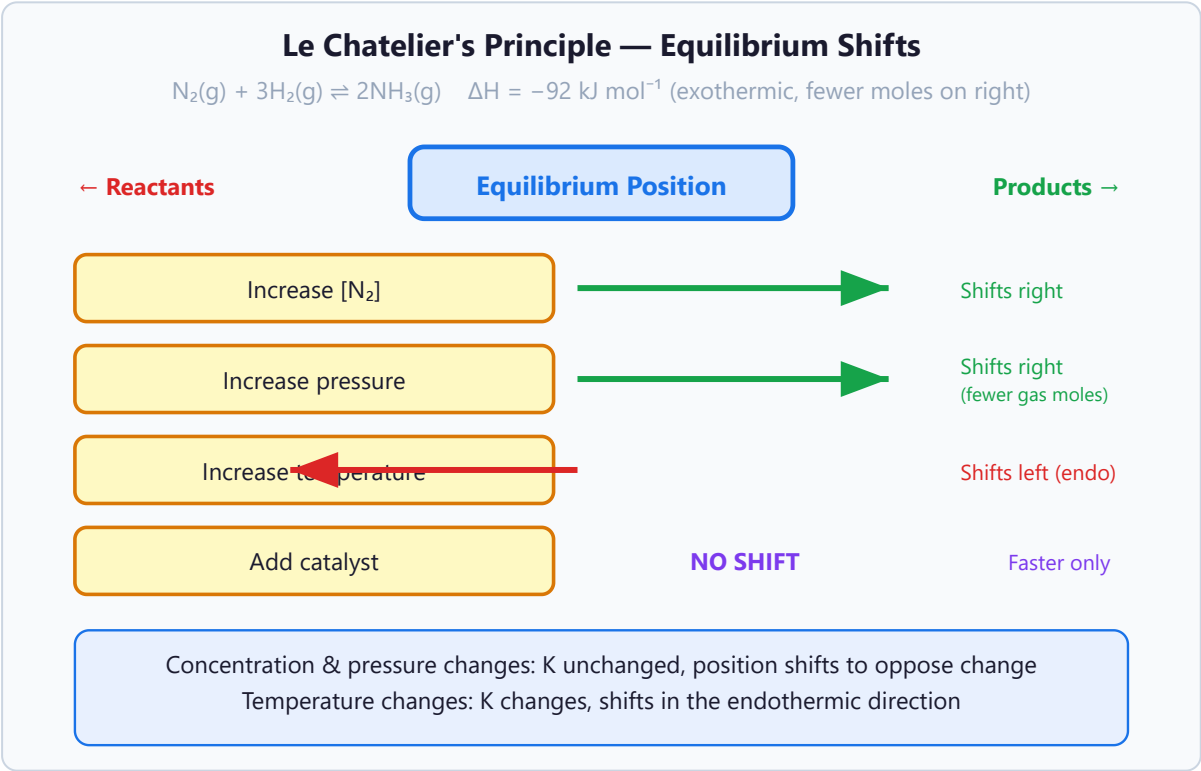
Increase pressure	Shifts towards the side with fewer moles of gas	No change
Increase temperature	Shifts in the endothermic direction	Changes (increases if forward reaction is endothermic)
Add a catalyst	No change (speeds up both forward and reverse equally)	No change

Exam Tip: A catalyst does not change the position of equilibrium or the value of K. It only increases the rate at which equilibrium is reached. Temperature is the only factor that changes the value of the equilibrium constant.



3.2 Equilibrium Constants

The Equilibrium Constant K_c

For a general reaction: $aA + bB \rightleftharpoons cC + dD$

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

where square brackets denote equilibrium concentrations in mol dm^{-3}. K_c is constant at a given temperature. A large K_c means the equilibrium lies to the right (products favoured); a small K_c means it lies to the left (reactants favoured).

Only species in the aqueous or gaseous phase are included in the expression. Pure solids and pure liquids are excluded because their concentrations are effectively constant.

The Equilibrium Constant K_p

For gaseous equilibria, K_p is expressed in terms of partial pressures:

$$K_p = p_C{}^c \cdot p_D{}^d / (p_A{}^a \cdot p_B{}^b)$$

Partial pressure: The pressure that a gas would exert if it alone occupied the total volume of the mixture at the same temperature. The partial pressure of gas A is: $p_A = x_A \times P_{total}$, where x_A is the mole fraction of A.

Mole fraction: x_A = moles of A / total moles of all gases.

The total pressure equals the sum of all partial pressures (Dalton's law): $P_{total} = p_A + p_B + p_C + ...$

Exam Tip: When calculating K_p, pay careful attention to units. The units of K_p depend on the stoichiometry of the reaction. Work out the overall power of pressure in the numerator minus the denominator to determine the unit (e.g. Pa, kPa, atm, or dimensionless).

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Equilibrium Constants — What K Tells Us

K « 1
Reactants favoured

K ≈ 1
Neither side favoured

K » 1
Products favoured

K^c (concentrations)

aA + bB ⇌ cC + dD

$K^c = [C]^c[D]^d / [A]^A[B]^b$

K^p (partial pressures)

For gaseous equilibria

$K^p = p^{cc}\, p^{dd} / p^{AA}\, p^{bb}$

What affects the value of K?

✓ Temperature changes K (shifts equilibrium in endothermic direction)

✗ Concentration, pressure, and catalysts do NOT change K

3.3 Acids and Bases

Brønsted-Lowry Theory

Brønsted-Lowry acid: A proton (H^+) donor.

Brønsted-Lowry base: A proton (H^+) acceptor.

Conjugate acid-base pair: Two species that differ by a single proton. When an acid donates a proton, it forms its conjugate base. When a base accepts a proton, it forms its conjugate acid.

Example: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$. Here HCl is the acid (donates H^+) and H_2O is the base (accepts H^+). Cl^- is the conjugate base of HCl, and H_3O^+ is the conjugate acid of H_2O.

Strong and Weak Acids

Strong acid: An acid that fully dissociates (ionises) in aqueous solution. Examples: HCl, H_2SO_4, HNO_3. For a strong monobasic acid, $[H^+]$ = [acid].

Weak acid: An acid that only partially dissociates in aqueous solution. An equilibrium exists between the undissociated acid and its ions. Examples: CH_3COOH, H_2CO_3.

pH Calculations

$$pH = -\log_{10}[H^+]$$

$$[H^+] = 10^{-pH}$$

Strong acid pH: For a strong monobasic acid at concentration c, $[H^+] = c$, so $pH = -\log c$.

Strong base pH: For a strong base like NaOH at concentration c, $[OH^-] = c$. Use K_w to find $[H^+]$: $[H^+] = K_w / [OH^-]$, then calculate pH.

The Acid Dissociation Constant K_a

For a weak acid $HA \rightleftharpoons H^+ + A^-$:

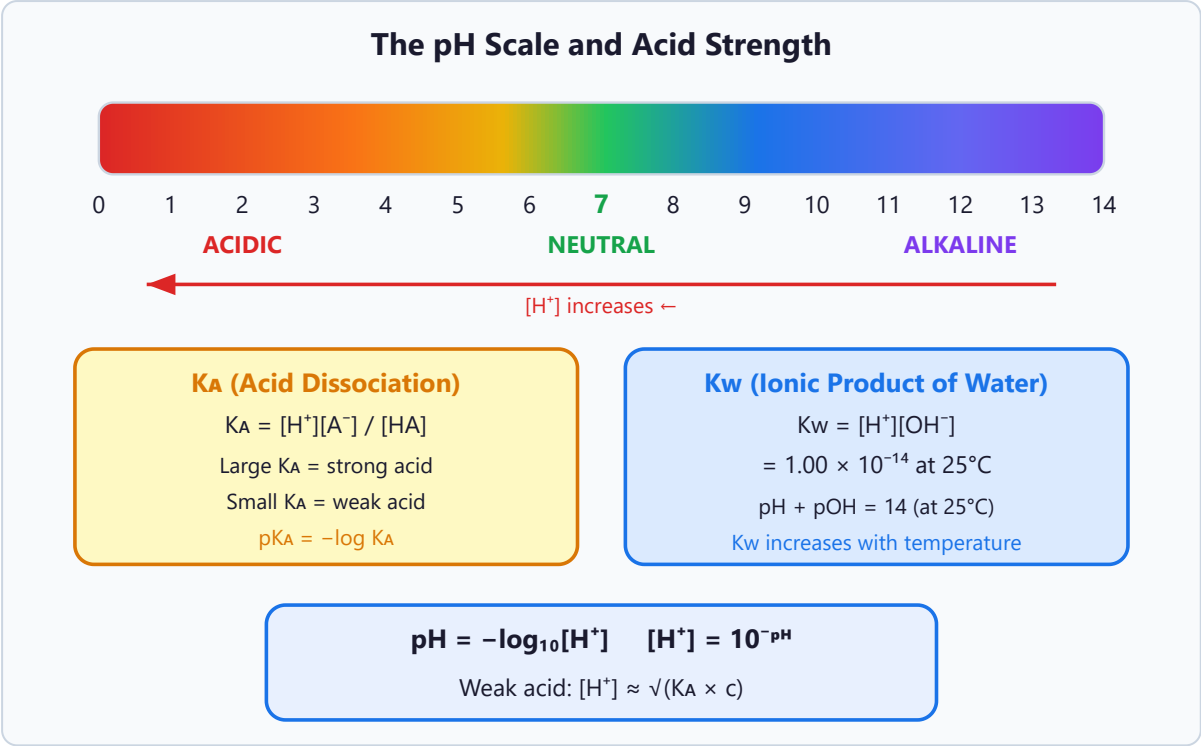
$$K_a = [H^+][A^-] / [HA]$$

For a weak acid, we assume $[H^+] = [A^-]$ (because the autoionisation of water contributes negligibly) and that the extent of dissociation is so small that the equilibrium concentration of HA is approximately equal to the initial concentration. Therefore:

$$[H^+] \approx \sqrt{(K_a \times c)}$$

$$pK_a = -\log_{10} K_a$$

The smaller the K_a (or larger the pK_a), the weaker the acid.



The Ionic Product of Water K_w

K_w: The ionic product of water. At 25 °C, $K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$ mol^2 dm^{-6}. K_w increases with temperature because the dissociation of water is endothermic.

In pure water at 25 °C: $[H^+] = [OH^-] = 1.00 \times 10^{-7}$ mol dm^{-3}, giving pH = 7.00. At higher temperatures, K_w is larger, so $[H^+]$ is greater and the pH of pure water is less than 7 — but the water is still neutral because $[H^+] = [OH^-]$.

3.4 Buffer Solutions

Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added. It consists of a weak acid and its conjugate base (or a weak base and its conjugate acid) in roughly equal concentrations.

How Buffers Work

An acidic buffer (e.g. ethanoic acid + sodium ethanoate) contains a large reservoir of both HA molecules and A^- ions:

- **When acid is added:** The extra H^+ ions react with the conjugate base A^- to form HA. This removes the added H^+, so the pH changes very little: $H^+ + A^- \rightarrow HA$
- **When base is added:** The extra OH^- ions react with the weak acid HA to form A^- and water. This removes the added OH^-: $OH^- + HA \rightarrow A^- + H_2O$

The buffer works because HA and A^- are present in large, roughly equal concentrations, so small additions of H^+ or OH^- cause only tiny changes to the ratio $[HA]/[A^-]$ and hence tiny changes in pH.

Calculating Buffer pH

Starting from $K_a = [H^+][A^-] / [HA]$, rearranging gives:

$$[H^+] = K_a \times [HA] / [A^-]$$

Then $pH = -\log[H^+]$. The Henderson-Hasselbalch equation provides an equivalent form:

$$pH = pK_a + \log_{10}([A^-] / [HA])$$

When $[HA] = [A^-]$, $pH = pK_a$. This is also the point at which the buffer has maximum buffering capacity. When choosing a buffer for a particular pH, select a weak acid whose pK_a is close to the desired pH.

Exam Tip: In buffer calculations, you can use moles instead of concentrations for [HA] and $[A^-]$ because they are in the same solution volume, so the volume cancels in the ratio. Be careful when the buffer is made by adding a strong base to an excess of weak acid — work out the moles of HA remaining and A^- formed.

Worked Example — pH of a Weak Acid:

Calculate the pH of 0.100 mol dm^{-3} ethanoic acid ($K_a = 1.74 \times 10^{-5}$ mol dm^{-3}).

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Assumptions: $[H^+] = [A^-]$; $[HA]_{eq} \approx [HA]_{initial}$

$[H^+] = \sqrt{(K_a \times [HA])} = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = \sqrt{(1.74 \times 10^{-6})} = 1.32 \times 10^{-3}$ mol dm^{-3}

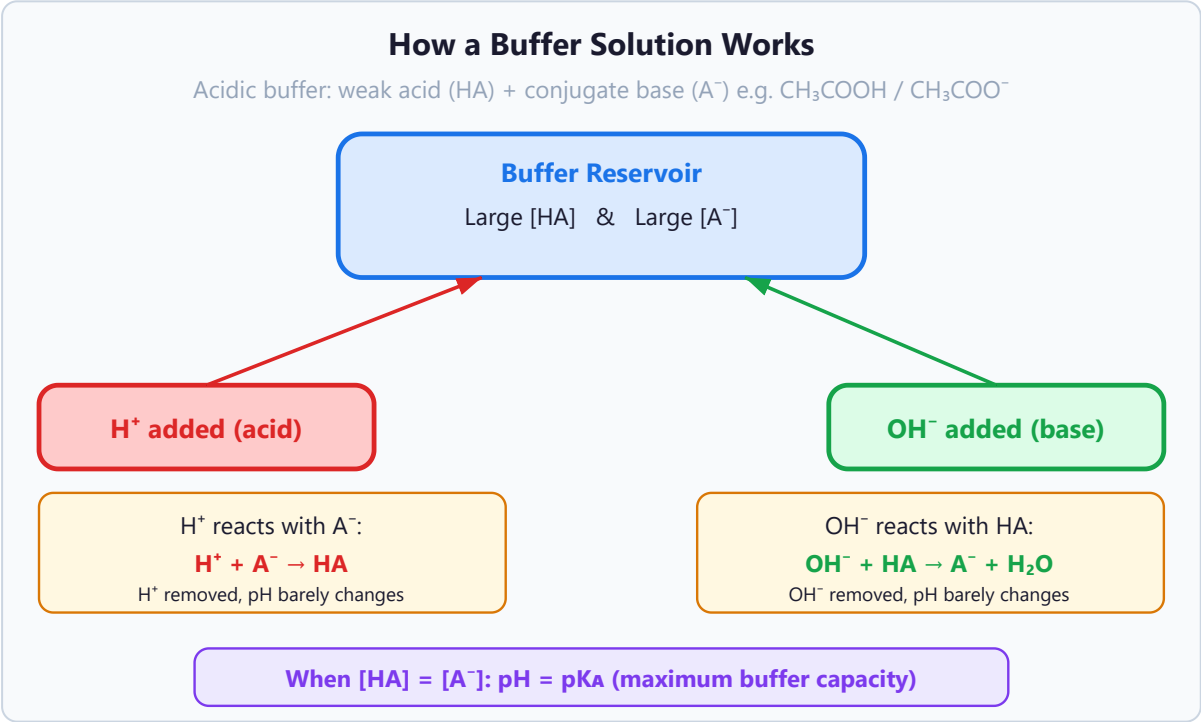
pH = $-\log(1.32 \times 10^{-3})$ = **2.88**

Worked Example — Buffer pH:

A buffer is made from 0.200 mol CH_3COOH and 0.100 mol CH_3COONa. $K_a = 1.74 \times 10^{-5}$.

$[H^+] = K_a \times [HA] / [A^-] = 1.74 \times 10^{-5} \times 0.200 / 0.100 = 3.48 \times 10^{-5}$ mol dm^{-3}

pH = $-\log(3.48 \times 10^{-5})$ = **4.46**



3.5 Titration Curves and Indicators

Titration Curves

A pH titration curve plots pH against volume of reagent added. The shape depends on the strengths of the acid and base:

Combination	pH at equivalence point	Suitable indicator
Strong acid + strong base	7	Any indicator with a range between pH 3–11 (e.g. phenolphthalein, methyl orange)
Strong acid + weak base	< 7 (acidic)	Methyl orange (range pH 3.1–4.4)
Weak acid + strong base	> 7 (basic)	Phenolphthalein (range pH 8.3–10.0)
Weak acid + weak base	≈ 7 (variable)	No suitable indicator (gradual pH change, no sharp equivalence point)

Key features of a strong acid–strong base curve: the pH changes slowly initially, then very sharply near the equivalence point (a near-vertical section), then levels off. The equivalence point is at the centre of the vertical section.

For a weak acid–strong base titration: the initial pH is higher (because the acid is weak), the curve rises more gradually, and the equivalence point pH is above 7 because the salt formed is slightly basic (the conjugate base of the weak acid hydrolyses).

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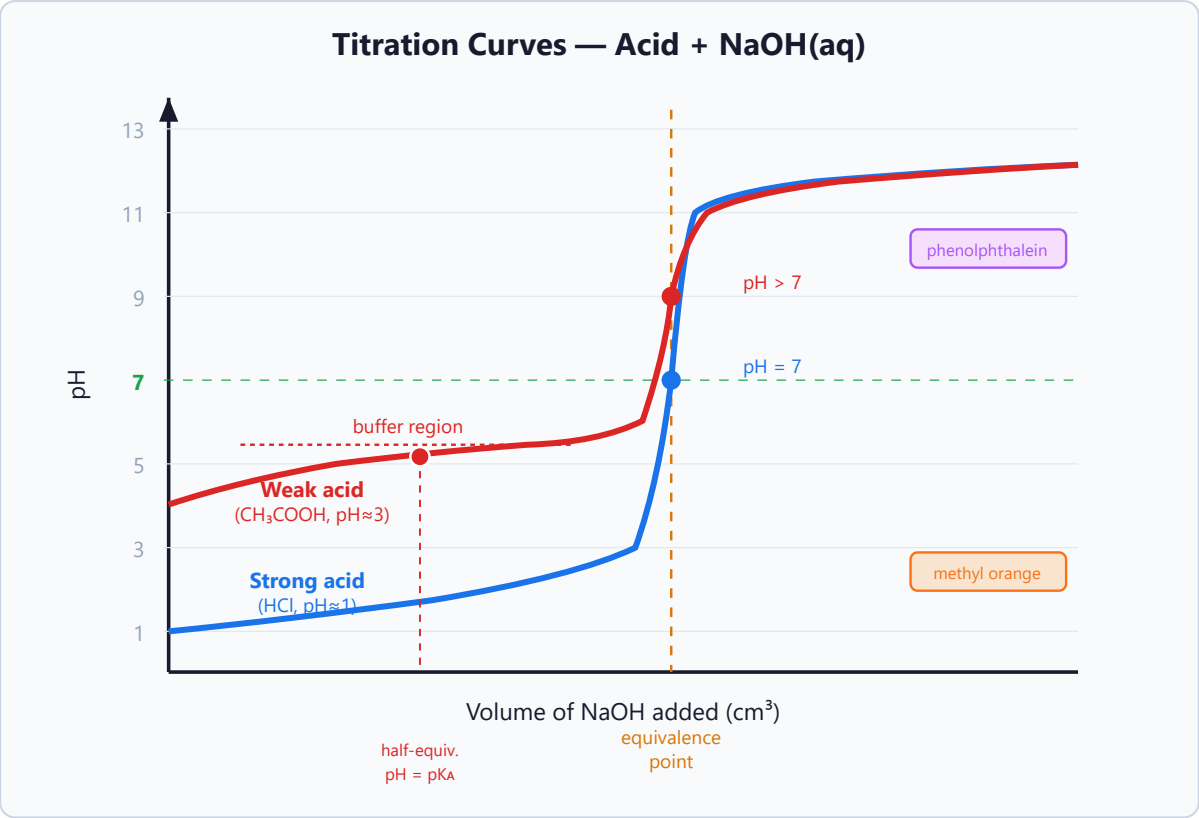

The Half-Equivalence Point

At the half-equivalence point (half the volume needed to reach the equivalence point), exactly half the weak acid has been neutralised, so [HA] = [A^-]. At this point, pH = pK_a. This provides an experimental method for determining K_a.

Indicators

Indicator: A weak acid (HIn) whose conjugate base (In^-) has a different colour. The indicator changes colour over a range of about 2 pH units centred on its pK_{In} value. A suitable indicator must have its colour change range within the steep (vertical) portion of the titration curve.

The indicator equilibrium: HIn(aq) ⇌ H^+(aq) + In^-(aq). In acidic solution [HIn] dominates (colour A); in basic solution [In^-] dominates (colour B). The colour change occurs at pH ≈ pK_{In}.

Neutralisation

Neutralisation is the reaction between an acid and a base to form a salt and water. For strong acid + strong base: H^+(aq) + OH^-(aq) → H_2O(l). The standard enthalpy of neutralisation for this reaction is

approximately -57.1 kJ mol^{-1}. For reactions involving weak acids or bases, the enthalpy of neutralisation is less exothermic because some energy is used to complete the dissociation of the weak acid or base.

Exam Tip: When sketching titration curves, label the axes clearly (pH on y-axis, volume of titrant on x-axis). Mark the equivalence point, the initial pH, and the buffer region (the relatively flat section before the equivalence point in a weak acid–strong base titration). The buffer region is where pH changes least per unit volume of titrant added.

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